

Model V0b — AR(2) Only

```
library('rjags')

## Loading required package: coda
## Linked to JAGS 4.3.2
## Loaded modules: basemod,bugs

library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(tibble)
library(stringr)
library(ggplot2)
library(tidyr)

output_folder = "../data/models/v0b/"
save_output   = TRUE
if (save_output) dir.create(output_folder, recursive = TRUE, showWarnings = FALSE)
```

Preprocess

Load data Loading data in as a wide data frame

With ~ rows (regions) x columns (weeks)

1st column is the region names

```
uecDf      = read.csv('../data/wide_weekly_scaledPer10k.csv')
regions    = uecDf$Region # saving region names
n.region   = length(uecDf$Region) # number of regions
uecDf      = uecDf %>% select(-Region) # remove region col
n.weeks    = length(names(uecDf))
head(uecDf)
```

```
##   X2023.01.01 X2023.01.08 X2023.01.15 X2023.01.22 X2023.01.29 X2023.02.05
## 1   0.9001159   6.857584   5.521329   5.289341   4.222193   4.491300
## 2   0.6823454   4.953323   3.285367   2.426117   2.021764   1.726924
## 3   0.6590512   6.013842   4.520679   3.604186   3.738056   3.233470
## 4   1.1620616   6.923950   3.776700   4.599827   3.921958   5.326116
## 5   1.0666825   7.142722   4.982352   5.022859   5.914012   5.171385
## 6   1.2769005  10.531138   6.002749   7.937845   6.489814   6.081732
```

##	X2023.02.12	X2023.02.19	X2023.02.26	X2023.03.05	X2023.03.12	X2023.03.19
## 1	5.252223	5.632684	6.170898	5.577007	7.599948	6.412166
## 2	1.760620	3.479119	3.462271	2.485085	2.089156	2.156549
## 3	4.325023	4.170558	3.604186	4.252940	4.345619	3.923414
## 4	6.439758	7.287095	7.819706	7.577610	7.432352	7.359723
## 5	3.807651	5.900510	5.860003	5.009357	7.412768	6.170556
## 6	5.699978	7.793042	7.977337	6.410830	7.635075	8.688189
##	X2023.03.26	X2023.04.02	X2023.04.09	X2023.04.16	X2023.04.23	X2023.04.30
## 1	6.996777	6.653434	4.546977	5.215105	5.725479	5.753318
## 2	2.746230	2.897862	2.232365	1.389963	1.920676	1.819588
## 3	4.613358	3.799842	2.018344	3.212875	4.283833	4.716335
## 4	8.715462	8.328108	8.037593	6.996579	8.110222	8.400737
## 5	6.373090	7.048206	5.846500	6.211063	6.589127	6.008528
## 6	9.543844	8.806664	9.267401	8.477566	8.082648	7.319141
##	X2023.05.07	X2023.05.14	X2023.05.21	X2023.05.28	X2023.06.04	X2023.06.11
## 1	6.736950	7.405077	4.825364	4.639773	4.194355	3.2385613
## 2	2.661990	2.678838	2.325029	2.737806	1.710076	0.5307131
## 3	5.272410	4.386810	2.924540	3.367340	1.966856	2.1522141
## 4	8.884929	7.504981	7.311304	8.763881	7.747077	7.9407542
## 5	5.887007	6.332584	6.062537	5.198389	3.564610	2.4574205
## 6	8.819828	8.872484	7.740386	7.121682	5.805290	6.7662561
##	X2023.06.18	X2023.06.25	X2023.07.02	X2023.07.09	X2023.07.16	X2023.07.23
## 1	4.482020	4.6212136	5.1872659	5.1315886	4.6768909	2.459080
## 2	0.463321	0.9182179	0.4212009	0.2695686	0.8087057	0.943490
## 3	2.728884	2.1522141	1.9874513	2.7288838	2.3375722	2.265488
## 4	7.529191	8.1344312	9.5385889	8.8849293	8.5459946	8.473366
## 5	3.888665	3.7806469	2.7004620	2.1198627	3.5376053	3.821154
## 6	7.095354	7.9115174	5.8184537	4.3177666	5.8842733	5.884273
##	X2023.07.30	X2023.08.06	X2023.08.13	X2023.08.20	X2023.08.27	X2023.09.03
## 1	2.7189068	2.7467454	2.820982	4.194355	3.591184	3.841732
## 2	0.7750096	0.7076175	2.004916	2.013340	1.718500	1.297299
## 3	3.1098978	2.9863257	2.605312	2.574419	2.687693	2.203702
## 4	9.1754447	7.9407542	7.771287	7.093418	7.553400	8.521785
## 5	4.5097716	4.3612462	2.497927	3.834656	5.846500	6.926685
## 6	6.2002075	6.9505510	7.845698	6.687273	6.331847	6.766256
##	X2023.09.10	X2023.09.17	X2023.09.24	X2023.10.01	X2023.10.08	X2023.10.15
## 1	4.695450	4.584095	4.435623	5.632684	5.512050	5.706920
## 2	1.575291	1.389963	1.861708	2.333453	2.316605	2.569325
## 3	2.286084	2.450847	3.593889	2.873051	2.842158	3.130493
## 4	9.804895	9.151235	9.320702	10.143829	10.095410	8.739672
## 5	5.954519	4.563781	5.319910	6.616132	5.535947	5.468436
## 6	7.924681	6.239699	5.410372	5.489356	6.134388	6.015913
##	X2023.10.22	X2023.10.29	X2023.11.05	X2023.11.12	X2023.11.19	X2023.11.26
## 1	4.491300	5.066632	6.013145	6.690552	5.911070	4.546977
## 2	2.139701	2.451389	2.855742	2.923134	3.041070	2.813622
## 3	2.924540	3.315851	3.089302	3.913116	3.738056	3.460019
## 4	9.441750	8.013383	6.923950	8.231270	9.659637	8.836510
## 5	4.860832	5.333413	5.981523	5.887007	6.251570	5.063366
## 6	5.397208	5.581503	5.805290	6.621453	7.253321	5.015454
##	X2023.12.03	X2023.12.10	X2023.12.17	X2023.12.24	X2023.12.31	X2024.01.07
## 1	4.407784	6.634875	6.653434	2.5611545	1.0857068	4.083000
## 2	2.392421	3.133735	2.771502	0.9856101	0.2021764	2.089156
## 3	4.057284	3.511507	3.892521	2.6156094	0.9061954	1.637330
## 4	8.231270	9.514379	10.095410	5.7618887	2.7841059	7.069208

## 5	6.062537	4.928343	6.602630	3.2540568	1.9713373	6.859174
## 6	5.120766	8.043157	8.990959	4.9891266	2.0535719	7.529764
##	X2024.01.14	X2024.01.21	X2024.01.28	X2024.02.04	X2024.02.11	X2024.02.18
## 1	3.953086	4.435623	5.140868	3.943807	4.714009	3.823173
## 2	3.083191	3.108463	2.939982	2.737806	3.378031	2.569325
## 3	2.677395	3.099600	2.924540	3.305554	4.294130	4.129368
## 4	7.698658	8.667043	10.192249	10.821699	12.129018	10.095410
## 5	6.481109	7.493782	7.439773	4.725809	4.253228	3.888665
## 6	6.318683	5.436700	6.805748	6.147552	6.726764	7.714059
##	X2024.02.25	X2024.03.03	X2024.03.10	X2024.03.17	X2024.03.24	X2024.03.31
## 1	4.584095	5.038793	4.751127	4.871761	3.999484	4.482020
## 2	3.285367	2.400845	2.333453	3.066343	2.712534	2.249213
## 3	2.811265	2.965730	3.109898	3.573293	2.780372	3.264363
## 4	9.369122	10.192249	9.635427	8.909139	10.168039	10.264877
## 5	3.402582	4.671799	3.605117	4.685302	5.076869	3.645624
## 6	6.239699	4.844323	4.146636	4.225619	5.568339	4.844323
##	X2024.04.07	X2024.04.14	X2024.04.21	X2024.04.28	X2024.05.05	X2024.05.12
## 1	4.751127	5.066632	4.537698	3.405593	3.507668	3.025132
## 2	2.164973	2.855742	2.948406	2.889438	2.181821	2.544053
## 3	3.923414	3.387935	3.696865	3.315851	2.831861	2.471442
## 4	8.763881	8.497575	9.732266	10.071200	7.892335	7.626029
## 5	5.306408	5.400924	3.281061	3.780647	5.076869	2.389909
## 6	6.739928	8.517058	6.042240	3.988668	3.909685	5.318225
##	X2024.05.19	X2024.05.26	X2024.06.02	X2024.06.09	X2024.06.16	X2024.06.23
## 1	4.342827	3.711818	4.500580	4.268591	3.841732	3.331357
## 2	2.965254	3.386455	2.080732	2.341877	2.822046	2.763078
## 3	2.121321	2.172809	1.657926	2.234595	2.347870	2.430251
## 4	10.434345	10.579602	10.216458	7.553400	7.335514	6.827112
## 5	2.551937	4.172214	4.037191	5.279403	4.887836	4.050693
## 6	6.516142	3.646407	4.883815	4.989127	7.582419	6.279191
##	X2024.06.30	X2024.07.07	X2024.07.14	X2024.07.21	X2024.07.28	X2024.08.04
## 1	3.071530	2.310607	2.394123	3.080809	3.544786	3.201443
## 2	2.973678	2.392421	1.432083	1.667956	1.710076	1.769044
## 3	2.780372	2.162512	2.502335	1.760902	1.688819	1.894772
## 4	8.788091	9.102816	11.233262	9.417541	7.747077	7.698658
## 5	3.348573	3.510601	4.523274	3.861661	4.266730	2.160370
## 6	6.055404	4.475734	5.699978	6.094896	5.910601	5.713142
##	X2024.08.11	X2024.08.18	X2024.08.25	X2024.09.01	X2024.09.08	X2024.09.15
## 1	3.108648	3.043691	2.431241	3.303518	2.718907	1.967264
## 2	1.819588	1.255179	1.802740	2.097580	2.392421	2.308181
## 3	2.327275	1.935963	2.584716	2.667098	2.718586	3.490912
## 4	9.030187	3.921958	0.000000	2.493591	4.793504	6.028195
## 5	3.186545	3.065024	2.349402	2.551937	2.713964	4.766316
## 6	5.555175	5.054946	4.673192	4.857487	4.752176	5.397208
##	X2024.09.22	X2024.09.29	X2024.10.06	X2024.10.13	X2024.10.20	X2024.10.27
## 1	3.238561	3.257120	3.841732	4.203634	3.182884	3.127207
## 2	2.232365	2.805198	3.344335	2.518781	2.619870	2.417693
## 3	3.326149	3.058409	3.614484	3.357042	3.398233	2.358168
## 4	6.754483	7.650239	9.417541	8.788091	8.086012	6.633435
## 5	4.712306	4.266730	5.360417	3.834656	3.051522	2.754471
## 6	6.634617	6.990043	6.739928	6.042240	5.568339	5.173422
##	X2024.11.03	X2024.11.10	X2024.11.17	X2024.11.24	X2024.12.01	X2024.12.08
## 1	2.988014	3.359195	3.758216	3.479830	4.519139	4.120118
## 2	2.754654	3.049494	2.510357	3.184279	4.009833	3.765536

## 3	2.172809	3.367340	2.780372	3.027516	4.201451	4.180856
## 4	9.030187	10.942747	9.635427	9.829104	8.715462	10.313297
## 5	2.970508	4.820325	3.186545	3.389080	5.009357	4.307237
## 6	4.673192	6.239699	6.502978	6.108060	5.739470	5.502520
##	X2024.12.15	X2024.12.22	X2024.12.29	X2025.01.05	X2025.01.12	X2025.01.19
## 1	4.816084	4.639773	1.503286	3.925248	5.428534	4.426343
## 2	3.597056	2.965254	1.120394	1.794316	3.217975	3.201127
## 3	3.810140	2.193405	1.019470	3.490912	3.470316	3.655675
## 4	8.763881	10.192249	4.987181	10.579602	11.765874	11.160633
## 5	3.537605	4.833827	1.741798	5.292906	7.264243	7.021201
## 6	5.660487	5.278733	2.566965	7.174338	8.003665	8.227452
##	X2025.01.26	X2025.02.02	X2025.02.09	X2025.02.16	X2025.02.23	X2025.03.02
## 1	3.544786	4.361386	3.971645	4.444902	4.482020	4.500580
## 2	3.504391	4.085649	4.523698	3.807656	3.748688	2.577749
## 3	5.045861	3.851330	4.242642	2.512633	3.521805	2.625907
## 4	8.521785	9.514379	9.223864	8.449156	10.385925	9.974362
## 5	6.494611	6.332584	4.914841	4.212721	4.509772	4.725809
## 6	8.477566	9.635991	9.754467	7.213829	7.450780	8.477566
##	X2025.03.09	X2025.03.16	X2025.03.23	X2025.03.30	X2025.04.06	X2025.04.13
## 1	3.600464	3.117927	2.635391	3.368475	2.440520	3.340636
## 2	1.785892	1.912252	1.069850	2.603022	3.066343	1.482627
## 3	3.027516	2.306679	1.946261	1.348995	1.606437	1.657926
## 4	8.884929	7.892335	6.270291	9.320702	7.722868	7.601820
## 5	3.497098	2.092858	3.834656	4.739311	4.104702	2.308895
## 6	6.608289	6.213371	4.041324	5.844782	4.936471	4.436242
##	X2025.04.20	X2025.04.27	X2025.05.04	X2025.05.11	X2025.05.18	X2025.05.25
## 1	3.962366	3.860291	2.421961	2.394123	2.691068	3.071530
## 2	2.367149	1.272027	1.339419	1.196211	1.010882	1.516323
## 3	1.668223	1.935963	1.781498	2.008047	1.832986	2.399358
## 4	8.739672	9.611218	8.400737	8.497575	9.248074	8.521785
## 5	2.281890	2.646453	2.106360	2.673457	2.362904	2.200877
## 6	4.778504	5.515683	4.594209	5.423536	6.266027	4.660029
##	X2025.06.01	X2025.06.08	X2025.06.15	X2025.06.22	X2025.06.29	X2025.07.06
## 1	2.783864	1.679598	2.310607	3.683980	2.728186	2.635391
## 2	1.828012	1.945948	1.760620	1.566867	1.263603	1.592139
## 3	2.121321	2.275786	1.585842	1.802093	1.853581	1.740307
## 4	8.473366	8.134431	8.667043	6.585016	9.659637	8.933349
## 5	1.579770	2.106360	1.755300	2.470923	3.429587	2.970508
## 6	5.792126	6.121224	6.423994	5.910601	6.410830	5.607831
##	X2025.07.13	X2025.07.20	X2025.07.27	X2025.08.03	X2025.08.10	X2025.08.17
## 1	1.744555	1.985823	1.818791	2.022941	1.865189	2.524036
## 2	1.499475	1.524747	1.693228	0.951914	1.330995	1.853284
## 3	2.069833	2.471442	2.162512	1.905070	1.997749	2.337572
## 4	6.657645	7.795497	8.594414	9.441750	6.778693	5.568212
## 5	3.335071	2.470923	3.254057	3.159541	2.308895	2.943504
## 6	4.594209	7.069026	7.134846	6.937387	6.094896	7.740386
##	X2025.08.24	X2025.08.31	X2025.09.07	X2025.09.14	X2025.09.21	X2025.09.28
## 1	1.429050	2.570434	2.533316	1.800232	1.967264	1.707436
## 2	1.861708	1.625835	1.533171	1.743772	1.769044	1.886980
## 3	2.842158	2.564121	2.368465	3.243768	3.573293	2.419954
## 4	9.369122	9.199654	9.272283	7.964964	8.884929	9.320702
## 5	2.700462	3.591615	4.644795	3.132536	3.146038	4.253228
## 6	8.240615	7.951009	7.134846	8.582877	7.253321	4.080816
##	X2025.10.05	X2025.10.12	X2025.10.19	X2025.10.26	X2025.11.02	X2025.11.09

```
## 1    2.421961    2.468359    1.735275    1.716716    2.273489    2.746745
## 2    2.375573    1.642683    1.246755    1.280451    1.457355    2.063884
## 3    2.131619    2.770075    2.955433    2.862754    2.728884    4.458893
## 4    10.700651    6.972370    4.502989    3.728281    4.260893    4.236683
## 5    4.536776    5.211892    5.670970    4.320739    5.076869    6.197560
## 6    5.291897    6.884731    4.778504    5.568339    4.804832    6.476650
##      X2025.11.16
## 1    2.895218
## 2    1.996492
## 3    3.665972
## 4    4.067216
## 5    2.808481
## 6    5.620995
```

Reformat Dataframe to Matrix Formatting as matrix with ~ rows (regions) x columns (weeks)

```
uecMat = as.matrix(uecDf)
```

Model V0b Specification

Baseline AR(2) model with no seasonality, no event effects, no covariates. $\mu_{i,t} = \alpha[i]$ (constant intercept per region). Two autoregressive lags capture temporal dependence. This tests whether a second lag improves on V0a.

Note: ϕ_1 and ϕ_2 use $\text{dunif}(-1,1)$ priors for consistency with the AR(1) prior in V0a/V1. This does not enforce the full AR(2) stationarity triangle ($\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, $|\phi_2| < 1$), but allows the posterior to signal non-stationarity if present in the data.

```
model =
"
model{
  for(i in 1:I){
    #-----
    #Likelihood
    #-----
    y[i,1] ~ dnorm(alpha[i], tau[i])
    y[i,2] ~ dnorm(alpha[i] + phi1 * (y[i,1] - alpha[i]), tau[i])
    for(t in 3:T){
      y[i,t] ~ dnorm(alpha[i]
                     + phi1 * (y[i,t-1] - alpha[i])
                     + phi2 * (y[i,t-2] - alpha[i]), tau[i])
    }
    #Full conditional mean used in the likelihood
    fullmod[i,1] <- alpha[i]
    fullmod[i,2] <- alpha[i] + phi1 * (y[i,1] - alpha[i])
    for(t in 3:T){
      fullmod[i,t] <- alpha[i]
                     + phi1 * (y[i,t-1] - alpha[i])
                     + phi2 * (y[i,t-2] - alpha[i])
    }
    #Residuals
    for(t in 1:T){
      resid[i,t] <- y[i,t] - fullmod[i,t]
    }
    #Uninformative Priors
```

```

    alpha[i] ~ dnorm(0, 0.001)
    tau[i] ~ dgamma(0.001, 0.001)
  }
  # dunif(-1,1) for consistency with V0a/V1; does not enforce
  # the full AR(2) stationarity region - see note above.
  phi1 ~ dunif(-1, 1)
  phi2 ~ dunif(-1, 1)
}
"

```

Fit

```

v0b_jags = jags.model(
  textConnection(model),
  list(
    y = uecMat,
    I = n.region,
    T = n.weeks
  ),
  n.chains = 4
)

## Compiling model graph
##   Resolving undeclared variables
##   Allocating nodes
## Graph information:
##   Observed stochastic nodes: 906
##   Unobserved stochastic nodes: 14
##   Total graph size: 4968
##
## Initializing model

update(v0b_jags, n.iter = 10000)

model_parameters = c("alpha", "tau", "phi1", "phi2")

v0b_sim = coda.samples(
  model = v0b_jags,
  variable.names = model_parameters,
  n.iter = 20000
)

```

```

dic_v0b = dic.samples(v0b_jags, n.iter = 20000)
dic_v0b

```

DIC

```

## Mean deviance: 2666
## penalty 14.31
## Penalized deviance: 2680

df_dic = data.frame(
  deviance = sum(dic_v0b$deviance),
  penalty = sum(dic_v0b$penalty),

```

```

    DIC      = sum(dic_v0b$deviance) + sum(dic_v0b$penalty)
  )
  if (save_output) write.csv(df_dic, paste0(output_folder, "dic.csv"), row.names = FALSE)

```

```

gelman_results = gelman.diag(v0b_sim, multivariate = FALSE)
gelman_results

```

Gelman Rubin Statistic

```
## Potential scale reduction factors:
```

```
##
```

```
##      Point est. Upper C.I.
```

```
## alpha[1]      1      1
```

```
## alpha[2]      1      1
```

```
## alpha[3]      1      1
```

```
## alpha[4]      1      1
```

```
## alpha[5]      1      1
```

```
## alpha[6]      1      1
```

```
## phi1          1      1
```

```
## phi2          1      1
```

```
## tau[1]        1      1
```

```
## tau[2]        1      1
```

```
## tau[3]        1      1
```

```
## tau[4]        1      1
```

```
## tau[5]        1      1
```

```
## tau[6]        1      1
```

```

if (save_output) {
  write.csv(
    as.data.frame(gelman_results$psrf) %>% rownames_to_column("param"),
    paste0(output_folder, "gelman.csv"),
    row.names = FALSE
  )
}

```

```

if (save_output) {
  samples_with_chain = do.call(rbind, lapply(seq_along(v0b_sim), function(ch) {
    df = as.data.frame(v0b_sim[[ch]])
    df$chain = ch
    df
  }))
  write.csv(samples_with_chain,
    paste0(output_folder, "raw_samples.csv"),
    row.names = FALSE)
}

```

Save